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Geometry

An Original Olympiad Practice Set

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Original content. These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original synthetic plane-geometry problems for olympiad training, moving from quick angle-chasing warm-ups to power-of-a-point, similarity, Ptolemy, and a coordinate/orthocenter finale. Each rewards a single clean insight rather than mechanical computation, and every numeric answer has been independently verified. Every closed-form answer was checked symbolically/numerically. Free to share for study.

Problems

1. DIFFICULTY 1/5 · ANGLE CHASING, ISOSCELES TRIANGLES, EXTERIOR ANGLE

In triangle ABC we have $AB = AC$ and $\angle BAC = 40^\circ$. The point D lies on ray BC beyond C so that $CD = CA$. Find $\angle ADB$.

2. DIFFICULTY 2/5 · POWER OF A POINT, TANGENT-SECANT, INTERSECTING CHORDS

A point P lies outside a circle ω. A line through P is tangent to ω at T. A second line through P meets ω at A and B, with A between P and B; here $PA = 4$ and $AB = 5$. A third line through P meets ω at C and D, with C between P and D, and $PC = 3$. Find PT and CD.

3. DIFFICULTY 3/5 · SIMILAR TRIANGLES, ALTITUDE TO HYPOTENUSE, INRADIUS OF RIGHT TRIANGLE

In right triangle ABC the right angle is at C, and $CA = 3$, $CB = 4$. The foot of the altitude from C to the hypotenuse AB is H. Let r_1 be the inradius of triangle ACH and r_2 the inradius of triangle BCH. Find the ratio $r_1 : r_2$, and prove it does not depend on the particular lengths but only on the legs.

4. DIFFICULTY 3/5 · INCIRCLE, EQUAL TANGENT LENGTHS, HERON'S FORMULA

The incircle of triangle ABC touches side BC at X, with $BX = 4$ and $CX = 6$. The tangent length from vertex A to the incircle equals 5 (that is, the incircle touches AB and AC at points whose distance to A is 5). Find the area of triangle ABC.

5. DIFFICULTY 3/5 · PTOLEMY'S THEOREM, CYCLIC QUADRILATERAL, EQUILATERAL TRIANGLE

Let ABC be an equilateral triangle inscribed in a circle, and let P be a point on the arc BC not containing A. Prove that $PA = PB + PC$.

6. DIFFICULTY 4/5 · ANGLE CHASING, CYCLIC QUADRILATERALS, DIRECTED ANGLES

Two circles ω_1 and ω_2 intersect at points P and Q. A line through P meets ω_1 again at A and ω_2 again at B. A second line through Q meets ω_1 again at C and ω_2 again at D. Prove that $AC \parallel BD$.

7. DIFFICULTY 4/5 · POWER OF A POINT, SIMILAR TRIANGLES, ALTITUDE ON HYPOTENUSE, RIGHT ANGLES

Let AB be a diameter of a circle ω with center O, and let P be a point outside ω on the line AB extended beyond B. From P draw a tangent PT to ω (with T the point of tangency), and let M be the foot of the perpendicular from T to line AB. Prove that $PM \cdot PO = PT^2$, and deduce that M is the same point for both tangents from P.

8. DIFFICULTY 5/5 · ORTHOCENTER, REFLECTION, CYCLIC QUADRILATERAL, COORDINATES, ANGLE CHASING

Let H be the orthocenter of an acute triangle ABC, and let H' be the reflection of H over the line BC. Prove that H' lies on the circumcircle of ABC. Then, in the specific triangle with $A = (0, 5)$, $B = (-3, 0)$, $C = (4, 0)$, find the coordinates of H' and verify it lies on the circumcircle.

Solutions

1. Answer: 35°

Since $AB = AC$, the base angles satisfy $\angle ABC = \angle ACB = \frac{180^\circ - 40^\circ}{2} = 70^\circ$.\par Now look at triangle ACD. Because D lies on segment BC extended beyond C, the angle $\angle ACD$ is the supplement of $\angle ACB$:

$$\angle ACD = 180^\circ - 70^\circ = 110^\circ.$$

We are given $CD = CA$, so triangle ACD is isosceles with apex at C. Its two equal base angles are

$$\angle CAD = \angle ADC = \frac{180^\circ - 110^\circ}{2} = 35^\circ.$$

Finally $\angle ADB$ is the same angle as $\angle ADC$ (since B, C, D are collinear and B lies on the C-side), so $\angle ADB = 35^\circ$.

Second method. Exterior-angle view: in triangle ACD, the angle $\angle ACB = 70^\circ$ is exterior at C, hence equals the sum of the two remote interior angles $\angle CAD + \angle ADC$. With $CA = CD$ those two are equal, so each is $70^\circ/2 = 35^\circ$, giving $\angle ADB = 35^\circ$ immediately.

2. Answer: $PT = 6, \quad CD = 9$

The power of the point P with respect to ω is a single number that every line through P must reproduce.\par Using the secant through A, B:

$$\operatorname{pow}(P) = PA \cdot PB = PA\,(PA + AB) = 4 \cdot (4 + 5) = 4 \cdot 9 = 36.$$

For the tangent, $\operatorname{pow}(P) = PT^2$, so

$$PT = \sqrt{36} = 6.$$

For the secant through C, D:

$$PC \cdot PD = 36 \;\Rightarrow\; PD = \frac{36}{3} = 12,$$

hence $CD = PD - PC = 12 - 3 = 9$.

Second method. One can derive $PT^2 = PA \cdot PB$ from scratch: triangles PTA and PBT are similar because $\angle TPA = \angle BPT$ (shared) and $\angle PTA = \angle PBT$ (tangent-chord angle equals the inscribed angle subtending TA). Then $\frac{PT}{PB} = \frac{PA}{PT}$ gives $PT^2 = PA \cdot PB = 36$, and the same similarity logic with the C, D secant fixes PD.

3. Answer: $r_1 : r_2 = CA : CB = 3 : 4$
The altitude to the hypotenuse creates the classical chain of similar triangles

$$\triangle ACH \sim \triangle ABC \sim \triangle CBH,$$

because each shares an acute angle with $\triangle ABC$ and each has a right angle. In particular $\triangle ACH \sim \triangle CBH$.\par Now here is the key observation: the inradius is a length, and any length scales by the same factor under a similarity. So if $\triangle ACH \sim \triangle CBH$ with ratio k, then $r_1/r_2 = k$. To find k, compare corresponding hypotenuses: the hypotenuse of $\triangle ACH$ is CA, and the corresponding hypotenuse of $\triangle CBH$ is CB (these are the sides opposite the right angles at H). Hence

$$\frac{r_1}{r_2} = \frac{CA}{CB} = \frac{3}{4}.$$

This depends only on the legs, as claimed.\par As a concrete check with $CA = 3, CB = 4, AB = 5$: the altitude is $CH = \frac{CA \cdot CB}{AB} = \frac{12}{5}$, with $AH = \frac{CA^2}{AB} = \frac{9}{5}$ and $BH = \frac{CB^2}{AB} = \frac{16}{5}$. Using $r = \frac{\text{leg}_1 + \text{leg}_2 - \text{hyp}}{2}$ for a right triangle,

$$r_1 = \frac{CH + AH - CA}{2} = \frac{\frac{12}{5} + \frac{9}{5} - 3}{2} = \frac{3}{5}, \qquad r_2 = \frac{CH + BH - CB}{2} = \frac{\frac{12}{5} + \frac{16}{5} - 4}{2} = \frac{4}{5},$$

so $r_1 : r_2 = \frac{3}{5} : \frac{4}{5} = 3 : 4$, matching the synthetic argument.

Second method. Avoid coordinates entirely: in any similar figures the inradius is proportional to a chosen homologous side. The similarities $\triangle ACH \sim \triangle ABC$ and $\triangle CBH \sim \triangle ABC$ have ratios CA/AB and CB/AB respectively (matching hypotenuse to hypotenuse). Since the inradius of $\triangle ABC$ is a fixed length ρ, we get $r_1 = \rho \cdot CA/AB$ and $r_2 = \rho \cdot CB/AB$, so $r_1/r_2 = CA/CB$ directly.

4. Answer: $30\sqrt{2}$
Write the tangent lengths from the three vertices as x (from A), y (from B), z (from C). Equal tangents from an external point give

$$y = BX = 4, \qquad z = CX = 6, \qquad x = 5 \text{ (given)}.$$

The side lengths are then

$$a = BC = y + z = 10, \quad b = CA = z + x = 11, \quad c = AB = x + y = 9.$$

The semiperimeter is

$$s = x + y + z = 5 + 4 + 6 = 15.$$

(Notice the consistency $s - a = 15 - 10 = 5 = x$, exactly the A-tangent length, as it must be.) By Heron's formula,

$$[ABC] = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{15 \cdot 5 \cdot 4 \cdot 6} = \sqrt{1800} = 30\sqrt{2}.$$

Second method. Once r is found the area follows from $[ABC] = rs$. Here $r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}} = \sqrt{\frac{5 \cdot 4 \cdot 6}{15}} = \sqrt{8} = 2\sqrt{2}$, so $[ABC] = rs = 2\sqrt{2} \cdot 15 = 30\sqrt{2}$.

5. Answer: Proof

Since P lies on the circumcircle of ABC, the quadrilateral $ABPC$ (vertices in this cyclic order A, B, P, C around the circle) is a cyclic quadrilateral. Apply Ptolemy's theorem, which states that for a cyclic quadrilateral the product of the diagonals equals the sum of the products of the two pairs of opposite sides:

$$AP \cdot BC = AB \cdot PC + AC \cdot PB.$$

Here AP and BC are the diagonals of $ABPC$, while AB, PC and AC, PB are the two pairs of opposite sides.\par Because triangle ABC is equilateral, $AB = AC = BC =: \ell$. Substituting,

$$AP \cdot \ell = \ell \cdot PC + \ell \cdot PB.$$

Dividing by $\ell > 0$ yields

$$PA = PB + PC,$$

as required. $\blacksquare$\par (Equality, not merely $\leq$, is what makes this clean: Ptolemy gives an equation precisely because P is concyclic with A, B, C; for a non-cyclic point the inequality $PA \leq PB + PC$ would hold instead.)

Second method. Trigonometric form. Let the circumradius be R. For P on arc BC, the inscribed angles give $\angle BPA = \angle CPA = 60^\circ$ (each subtends a side of the equilateral triangle). By the extended law of sines in triangles ABP and ACP, the chords are $PB = 2R\sin\angle PAB$, $PC = 2R\sin\angle PAC$, and $PA = 2R\sin\angle ABP$. Using $\angle PAB + \angle PAC = 60^\circ$ and the sum-to-product identity $\sin u + \sin v = 2\sin\frac{u+v}{2}\cos\frac{u-v}{2}$ reduces $PB + PC$ to exactly PA. A direct coordinate check (side 1, P the arc midpoint) gives $PA = PB + PC = \frac{2}{\sqrt{3}}$, confirming the identity.

6. Answer: Proof

We chase angles using the common chord PQ, which links the two circles. Throughout we use directed angles between lines, taken modulo 180°, so the argument is independent of how A, B, C, D happen to be arranged.\par Work in ω_1, which passes through A, C, P, Q. Since these four points are concyclic, the inscribed-angle theorem for the chord CQ gives

$$\angle(CA, AP) = \angle(CQ, QP),$$

because $\angle CAP$ (at A) and $\angle CQP$ (at Q) are angles subtending the same chord CQ (no, CP) — more precisely, in directed-angle form, points A, C, P, Q concyclic means $\angle(AC, AP) = \angle(QC, QP)$.\par Work in ω_2, through B, D, P, Q. Likewise, concyclicity of B, D, P, Q gives

$$\angle(BD, BP) = \angle(QD, QP).$$

Now use the collinearities built into the construction. Points A, P, B lie on one line through P, so line $AP =$ line BP. Points C, Q, D lie on one line through Q, so line $QC =$ line QD. The second collinearity makes the two right-hand sides equal:

$$\angle(QC, QP) = \angle(QD, QP).$$

Chaining the three equalities,

$$\angle(AC, AP) = \angle(BD, BP) = \angle(BD, AP),$$

the last step using $AP = BP$ as lines. Thus AC and BD make equal directed angles with the common line AP. Equal directed angles with a common transversal mean the lines are parallel:

$$AC \parallel BD. \qquad \blacksquare$$

Second method. Undirected version, tracking supplements. In ω_1, $APQC$ cyclic gives $\angle PAC = \angle PQC$ (inscribed angles on arc PC, up to supplement). In ω_2, $BPQD$ cyclic gives $\angle PBD = \angle PQD$. Since C, Q, D are collinear, $\angle PQC$ and $\angle PQD$ are the same angle, so $\angle PAC = \angle PBD$. These are corresponding angles cut by the transversal line APB on lines AC and BD; equal corresponding angles force $AC \parallel BD$.

7. Answer: Proof

Set up the standard right-angle facts. Since PT is tangent at T, the radius $OT \perp PT$, so triangle OTP is right-angled at T. The point M is the foot of the perpendicular from T onto the hypotenuse OP of this right triangle (note M lies on line $AB = OP$).\par This is exactly the altitude-on-hypotenuse configuration. In a right triangle, the foot of the altitude from the right angle satisfies the well-known relation: a leg squared equals the product of the hypotenuse and the adjacent segment of the hypotenuse. Applying it to right triangle OTP with altitude TM:

$$PT^2 = PM \cdot PO,$$

since PT is the leg adjacent to the segment PM. (Equivalently, $\triangle PTM \sim \triangle POT$: they share angle P and each has a right angle, giving $\frac{PT}{PO} = \frac{PM}{PT}$, i.e. $PT^2 = PM \cdot PO$.) This proves the stated identity.\par Now deduce the independence from the choice of tangent. The power of P with respect to ω is

$$PT^2 = \text{pow}(P) = PA \cdot PB,$$

a quantity determined by P and ω alone, the same for both tangents PT and PT'. From the identity,

$$PM = \frac{PT^2}{PO} = \frac{\text{pow}(P)}{PO},$$

which likewise depends only on P and ω, not on which tangent we chose. Both feet M and M' lie on the fixed line AB, on the same side of P, at the identical distance PM from P; hence they coincide, $M = M'$. So both tangent points project to the same foot M on AB. ■

Second method. Coordinate confirmation. Put $O = (0, 0)$, radius 1, $A = (-1, 0)$, $B = (1, 0)$, and $P = (p, 0)$ with $p > 1$. The tangent point is $T = \left(\frac{1}{p}, \pm\frac{\sqrt{p^2-1}}{p}\right)$ (it lies on ω and satisfies $OT \perp PT$). Its foot on the x-axis is $M = \left(\frac{1}{p}, 0\right)$, the same x-coordinate for either sign, so M is independent of the tangent. Then $PM = p - \frac{1}{p}$ and $PO = p$, so $PM \cdot PO = p^2 - 1$; and $PT^2 = \left(p - \frac{1}{p}\right)^2 + \frac{p^2-1}{p^2} = p^2 - 1$. The two agree, confirming $PM \cdot PO = PT^2$.

8. Answer: $H' = \left(0,\ -\frac{12}{5}\right)$**; it lies on the circumcircle** $x^2 + y^2 - x - \frac{13}{5}y - 12 = 0$**.**
Synthetic proof. We show $\angle BH'C = 180^\circ - \angle BAC$, which places H' on the arc BC opposite A, i.e. on the circumcircle.\par First, the standard orthocenter fact $\angle BHC = 180^\circ - \angle BAC$. Proof by angle chasing: let E be the foot of the altitude from B (on AC) and F the foot from C (on AB). In quadrilateral $AFHE$, the angles at F and E are right angles, so the remaining two angles sum to 180°: $\angle FHE = 180^\circ - \angle A$. Since $\angle BHC = \angle FHE$ (vertical angles at H), we get $\angle BHC = 180^\circ - \angle A$.\par Reflection over line BC is an isometry fixing B and C, so it preserves the angle subtended at the reflected point:

$$\angle BH'C = \angle BHC = 180^\circ - \angle BAC.$$

Thus $\angle BH'C + \angle BAC = 180^\circ$. Since H is interior to the acute triangle, its reflection H' lies on the opposite side of line BC from A. The locus of points on that far side of chord BC which subtend an angle supplementary to $\angle BAC$ is precisely the far arc of the circumcircle of ABC. Therefore H' lies on the circumcircle. $\blacksquare$\par Concrete computation for $A = (0,5),\ B = (-3,0),\ C = (4,0)$. Side BC is the x-axis. The altitude from A is perpendicular to BC, hence the vertical line $x = 0$. The altitude from B is perpendicular to AC; the direction of AC is $C - A = (4,-5)$, so this altitude through $B = (-3,0)$ satisfies $4(x+3) - 5y = 0$. Intersecting with $x = 0$: $4(3) - 5y = 0$, so $y = \frac{12}{5}$. Thus

$$H = \left(0,\ \frac{12}{5}\right).$$

Reflecting over BC (the x-axis) flips the sign of y:

$$H' = \left(0,\ -\frac{12}{5}\right).$$

The circumcircle through A, B, C is $x^2 + y^2 + Dx + Ey + F = 0$; substituting the three vertices gives $D = -1,\ E = -\frac{13}{5},\ F = -12$, i.e. $x^2 + y^2 - x - \frac{13}{5}y - 12 = 0$. Testing $H' = (0, -\frac{12}{5})$:

$$0 + \frac{144}{25} - 0 - \frac{13}{5}\left(-\frac{12}{5}\right) - 12 = \frac{144}{25} + \frac{156}{25} - 12 = \frac{300}{25} - 12 = 12 - 12 = 0.$$

So H' satisfies the circle equation and indeed lies on the circumcircle.

Second method. Vector/reflection proof. Place the circumcenter O at the origin with circumradius R; then Euler's identity gives the orthocenter $H = A + B + C$. Let $A^* = -A$ be the antipode of A, which lies on the circle. The midpoint M of BC satisfies $2M = B + C = H - A = H + A^*$, so A^* is the reflection of H over the point M. Reflecting H over the line BC (rather than over the point M) produces a point at the same distance from O as A^*, namely R; hence $|OH'| = R$ and H' lies on the circumcircle. The explicit coordinates above confirm $|OH'| = R$ numerically.